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Master Documentation: UG Gender Policy Case Management System

1. Executive Summary & Project Scope

The University of Ghana (UG) Gender Policy Case Management System is an institutional-grade, interactive reference website and secure administrative portal. It is built to digitize the UG Gender Policy (2023) and facilitate the secure reporting, tracking, and resolution of gender-based grievances.

In-Scope

- **Interactive Policy Reference:** Digitizing the policy into searchable, navigable tabs.
- **Secure Incident Reporting:** A multi-step reporting wizard for formal/informal complaints with evidence uploads.
- **Administrative Case Management:** A secure EOB dashboard to track the 21-day adjudication window.
- **Policy & Analytics Management:** Tools to manage definitions and aggregate annual metrics.

Out-of-Scope

- External Legal Adjudication (e.g., Police, Courts).
 - Automated HR/SIS database integrations.
 - In-App Real-Time Chat (communication happens offline/via official email).
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2. User Profiles & User Stories

2.1 The University Community (Complainants)

Students, academic faculty, administrative staff, and interns.

- **US-01:** As a student, I want to easily search policy definitions so I understand my rights under the Gender Policy.
- **US-02:** As a complainant, I want to report an incident through a secure, guided form so my privacy is protected.
- **US-03:** As a complainant, I want to upload screenshots and audio evidence so the EOB has proof of the grievance.

- **US-04:** As a complainant, I want to receive an anonymous Reference ID so I can safely track my case.

2.2 EOB Case Officers (Investigators)

Dedicated personnel appointed by the EOB Secretariat.

- **US-05:** As an investigating officer, I want to be notified when a new case is assigned to me so I can meet the 7-day contact SLA.
- **US-06:** As an investigating officer, I want to add internal notes to a case file so I can keep a secure audit trail of my interviews.
- **US-07:** As an investigating officer, I want to see a visual timeline on my cases so I don't breach the 21-day adjudication deadline.

2.3 System Administrators (EOB Secretariat)

High-level administrative staff overseeing policy implementation.

- **US-08:** As a system admin, I want to view a dashboard of all active cases so I can assign them to available officers.
- **US-09:** As a system admin, I want to update policy definitions directly from the dashboard so the public portal remains accurate without developer intervention.
- **US-10:** As a system admin, I want to generate annual analytics so I can easily compile the Vice-Chancellor's Gender Equality Annual Report.

3. Core Features & UI Walkthrough

3.1 Public Portal

Interactive Policy Reference & Definitions Users can navigate the official policy principles and search through the 33 official terms in real-time.

[insert screenshot of the Public Definitions/Overview Page]

Secure Incident Reporting Wizard A 5-step form (Path, Details, Narrative, Evidence, Review) guiding users to submit grievances securely.

[insert screenshot of the Report Incident Wizard]

Submission Receipt Confirmation screen providing the Reference ID and the 7-day EOB contact guarantee.

[insert screenshot of the Submission Confirmation Screen]

3.2 Administrative Dashboard

Operational Home Dashboard High-level live metrics tracking active filings, open enquiries, and panel hearings.

[insert screenshot of the Admin Dashboard metrics]

Case Management Registry Secure data grid with advanced filtering (by offence, type) and SLA timeline warning bars.

[insert screenshot of the Admin Cases Registry]

Policy Terminology Manager CRUD interface allowing administrators to update the public definition registry.

[insert screenshot of the Admin Definitions Manager]

4. System Architecture & Data Flow

Data Flow Diagram (DFD)

Illustrates how information moves from the complainant, through the API, into secure storage, and to the EOB administrators.

```
Parse error on line 9:
...ackend]          DB[(PostgreSQL Database
-----^
Expecting 'SPACE', 'GRAPH', 'DIR', 'TAGEND', 'TAGSTART', 'UP',
'DOWN', 'subgraph', 'end', 'MINUS', '--', '==', 'STR', 'STYLE',
'LINKSTYLE', 'CLASSDEF', 'CLASS', 'CLICK', 'DEFAULT', 'NUM',
'PCT', 'COMMA', 'ALPHA', 'COLON', 'BRKT', 'DOT', 'PUNCTUATION',
'UNICODE_TEXT', 'PLUS', 'EQUALS', 'MULT', got 'PS'
```

End-to-End SLA Sequence Diagram

Highlights the critical 7-day notification SLA and 21-day adjudication timer.

sequenceDiagram participant C as Complainant participant Sys as System participant SA as System Admin participant Off as EOB Case Officer C->>Sys: Submit Formal Complaint activate Sys Sys->>Sys: Generate Reference ID & Save Sys-->>C: Return Success (Ref ID + 7-Day SLA) deactivate Sys Sys->>SA: Dashboard Alert - New Initial Review SA->>Sys: Review Case details activate Sys SA->>Sys: Assign EOB Case Officer Sys->>Sys: Start 21-Day Adjudication Timer Sys-->>Off: Notification - New Case Assigned deactivate Sys Note over Off, C: Must happen within 7 working days Off->>C: Formal Contact & Acknowledgement Off->>Sys: Update Case Status to Investigation Sys-->>Sys: Log Timestamp (Audit Trail)

5. Database Architecture

Entity Relationship Diagram (ERD)

```
Parse error on line 1:
erDiagram      USERS
^
Expecting 'NEWLINE', 'SPACE', 'GRAPH', got 'ALPHA'
```

Security Constraints:

- Strict Foreign Key integrity (**ON DELETE RESTRICT**) on Users.
- Encryption at rest for the PostgreSQL database and Evidence Blob Storage.
- Append-only immutability for **CASE_NOTES** and **EVIDENCE** to preserve legal chain of custody.

Data Dictionary

1. **users** Table

Stores authentication and profile data for EOB Secretariat staff and assigned Investigators.

- **role**: Enforces RBAC. **ADMIN** has global oversight, **OFFICER** has access only to assigned cases.
- **is_active**: Used for soft-deleting or revoking access for offboarded staff to preserve historical assignment data on closed cases.

2. **cases** Table (The Core Record)

Stores the primary details of a grievance submitted by the university community.

- **reference_id**: A unique, pseudo-anonymous string (e.g., **#GBC-24-0812**) returned to the user upon submission.
- **incident_type**: Restricted by an Enum that perfectly maps to the 9 official policy offences (e.g., **sexist_remarks**, **promotion_denial**, **gbv**).
- **assigned_officer_id**: Foreign Key to the **users** table. Setting this value starts the 7-day notification SLA.
- **submitted_at**: Timestamp used to calculate the 21-day adjudication deadline.

3. **case_notes** Table

Acts as an audit trail and investigation log for an active case.

- **visibility**: Determines if the note is purely for the internal officer (**internal**) or if it should be included in the dossier sent to the Adjudication Panel (**panel**).

4. **evidence** Table

Manages all uploaded media and documents related to a case.

- **storage_uri**: A secure URI pointing to the actual file in cloud blob storage (e.g., AWS S3).
- **uploaded_by_id**: Tracks provenance. If null, the evidence was uploaded by the anonymous complainant during the initial public submission. If populated, it was uploaded by an investigating officer.

5. **definitions** Table

Centralized registry for the policy terminology, served to the public **/definitions** route.

- Allows EOB Administrators to update definitions dynamically without requiring frontend code changes.

6. API Infrastructure

The system utilizes a secure, v1 RESTful API partitioned by functional domains to handle end-to-end data flow.

6.1 Public Content & Policy Reference

Endpoints serving the public-facing educational portals. No authentication required.

Method	Endpoint	Description	Query Parameters / Payload
GET	/api/v1/policy/definitions	Retrieves all official policy terms and acronyms.	? type=term acronym, ?search=query
GET	/api/v1/policy/offences	Retrieves the 9 official policy offences.	-

6.2 Complaint Intake & Reporting (Public)

Endpoints handling the ingestion of grievances from the university community.

Method	Endpoint	Description	Payload / Response
POST	/api/v1/complaints	Submits a new formal or informal grievance.	Body: { path, incidentType, incidentDate, location, parties, narrative, witnesses, evidenceUris[] }

Method	Endpoint	Description	Payload / Response
			Returns: { referenceId: "#GBC-24-0812" }
POST	/api/v1/complaints/evidence	Handles multipart/form-data for evidence uploads. Streams to secure blob storage.	Form-Data: file Returns: { uri: "s3://...", type: "video/mp4" }
GET	/api/v1/complaints/status/:refId	(Optional) Allows complainants to securely check the high-level status of their case using their Reference ID.	Returns: { status: "Investigation", daysRemaining: 14 }

6.3 Authentication & Authorization

Endpoints handling Role-Based Access Control (RBAC) for EOB Officers and Administrators.

Method	Endpoint	Description	Payload
POST	/api/v1/auth/login	Authenticates staff and issues secure HTTP-only JWTs.	Body: { email, password }
GET	/api/v1/auth/me	Retrieves the authenticated user's profile and RBAC role	-

Method	Endpoint	Description	Payload
		(ADMIN, OFFICER).	

6.4 Case Management (Admin & Officers)

Endpoints for the lifecycle management of submitted cases. Requires **Bearer <Token>**.

Method	Endpoint	Description	Query Parameters / Payload
GET	/api/v1/admin/cases	Lists cases based on access level. Admins see all; Officers see assigned cases.	?status=open, ?type=informal, ?offenceType=gbv
GET	/api/v1/admin/cases/:id	Retrieves full secure details of a specific grievance, including evidence URIs.	-
PATCH	/api/v1/admin/cases/:id	Updates operational properties of a case.	Body: { status, priority }
POST	/api/v1/admin/cases/:id/assign	Assigns an EOB Case Officer to a case.	Body: { officerId } (Requires ADMIN role)

Method	Endpoint	Description	Query Parameters / Payload
		Triggers the 7-day contact notification.	
GET	/api/v1/admin/cases/:id/notes	Retrieves internal investigation notes for a case.	-
POST	/api/v1/admin/cases/:id/notes	Appends a new internal investigation note/log to the case timeline.	Body: { noteText, visibility: "internal panel" }

6.5 Global Policy Management (Admin Only)

Endpoints for managing the underlying taxonomies of the policy. Requires **ADMIN** role.

Method	Endpoint	Description	Payload
POST	/api/v1/admin/definitions	Creates a new policy term or acronym.	Body: { term, definition, type }
PUT	/api/v1/admin/definitions/:id	Updates an existing definition.	Body: { term, definition }
DELETE	/api/v1/admin/definitions/:id	Removes a definition from the registry.	-

6.6 Reporting & Analytics (Admin Only)

Endpoints supporting the dashboard metrics and annual reporting requirements.
Requires **ADMIN** role.

Method	Endpoint	Description	Query Parameters
GET	/api/v1/admin/analytics/dashboard	Fetches real-time counts for active filings, open enquiries, and panel hearings.	-
GET	/api/v1/admin/analytics/annual	Fetches aggregated, anonymized metrics for the Gender Equality Annual Report.	?academicYear=2024-2025

6.7 System & User Management (Admin Only)

Endpoints for managing EOB staff access. Requires **ADMIN** role.

Method	Endpoint	Description	Payload
GET	/api/v1/admin/users	Lists all registered EOB Secretariat staff, Administrators, and Case Officers.	-
POST	/api/v1/admin/users	Provisions a new account for an EOB staff member.	Body: { name, email, role, department }

Method	Endpoint	Description	Payload
PUT	/api/v1/admin/users/:id	Updates user details or role.	Body: { name, role, isActive }